

1. 拉普拉斯逆變換

傅立葉變換與傅立葉逆變換是：

$$f(t) \xleftrightarrow[\frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega]{\int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt} F(\omega)$$

類似地，拉普拉斯逆變換 (Inverse Laplace Transform) 使用積分定義為：

$$f(t) = \mathcal{L}^{-1}\{F(s)\} = \frac{1}{2\pi i} \lim_{T \rightarrow \infty} \int_{\alpha-iT}^{\alpha+iT} F(s) e^{st} ds$$

但這是一個在複平面上的線積分，計算起來非常複雜。因此我們更偏向於湊出我們已知函數的拉普拉斯變換：

Question Find the inverse Laplace transform of following functions.

$$\begin{aligned} & \triangleright \mathcal{L}^{-1} \left\{ \frac{1}{s-3} + \frac{12}{(s-1)^4} + \frac{s+1}{s^2+9} + 5 \right\}. \\ & \quad \mathcal{L}^{-1} \left\{ \frac{1}{s-3} + \frac{12}{(s-1)^4} + \frac{s+1}{s^2+9} + 5 \right\} \\ & = \mathcal{L}^{-1} \left\{ \frac{1}{s} \Big|_{s \rightarrow s-3} + \frac{3!}{s^4} \Big|_{s \rightarrow s-1} + \frac{s}{s^2+3^2} + \frac{1}{3} \frac{3}{s^2+3^2} + 5 \cdot 1 \right\} \\ & = u(t) \cdot e^{3t} + t^3 e^t + \cos 3t + \frac{1}{3} \sin 3t + 5\delta(t) \end{aligned}$$

$$\begin{aligned} & \blacktriangleright \mathcal{L}^{-1} \left\{ \frac{1}{s^2+3s+2} + \frac{1}{s} e^{-s} + \frac{1}{s-2} e^{-3s} \right\}. \\ & \quad \mathcal{L}^{-1} \left\{ \frac{1}{s^2+3s+2} + \frac{1}{s} e^{-s} + \frac{1}{s-2} e^{-3s} \right\} \\ & = \mathcal{L}^{-1} \left\{ \left(\frac{1}{s+1} - \frac{1}{s+2} \right) + \frac{1}{s} e^{-s} + \frac{1}{s-2} e^{-3s} \right\} \\ & = e^{-t} - e^{-2t} + u(t-1) + e^{2(t-3)} u(t-3) \end{aligned}$$

1.1. 補充：部分分式分解

Case 1：相異一次式

若分式具有相異的一次因子：

$$F(s) = \frac{N(s)}{(s-a)(s-b)} = \frac{A}{s-a} + \frac{B}{s-b}$$

- 求 A ：把 A 的分母遮住，即計算 $A = (s-a)F(s) \Big|_{s=a}$
- 求 B ：把 B 的分母遮住，即計算 $B = (s-b)F(s) \Big|_{s=b}$

Question Decompose the following fraction as partial fraction.

$$\triangleright \frac{17x-53}{x^2-2x-15}.$$

$$\text{Let } \frac{17x-53}{x^2-2x-15} = \frac{17x-53}{(x-5)(x+3)} = \frac{A}{x-5} + \frac{B}{x+3},$$

$$\text{then } 17x-53 = A(x+3) + B(x-5).$$

$$\text{Therefore : } \begin{cases} A = \frac{17x-53}{x+3} \Big|_{x=5} = 4 \\ B = \frac{17x-53}{x-5} \Big|_{x=-3} = 13 \end{cases}$$

$$\text{Hence, } \frac{17x-53}{x^2-2x-15} = \frac{4}{x-5} + \frac{13}{x+3}.$$

$$\blacktriangleright \frac{-9x^2+4x+125}{(x-1)(x+3)(x+4)}.$$

$$\text{Let } \frac{-9x^2+4x+125}{(x-1)(x+3)(x+4)} = \frac{A}{x-1} + \frac{B}{x+3} + \frac{C}{x+4},$$

$$\text{then } -9x^2+4x+125 = A(x+3)(x+4) + B(x-1)(x+4) + C(x-1)(x+3).$$

$$\text{Therefore : } \begin{cases} A = \frac{-9x^2+4x+125}{(x+3)(x+4)} \Big|_{x=1} = 6 \\ B = \frac{-9x^2+4x+125}{(x-1)(x+4)} \Big|_{x=-3} = -8 \\ C = \frac{-9x^2+4x+125}{(x-1)(x+3)} \Big|_{x=-4} = -7 \end{cases}$$

$$\text{Hence, } \frac{-9x^2+4x+125}{(x-1)(x+3)(x+4)} = \frac{6}{x-1} - \frac{8}{x+3} - \frac{7}{x+4}.$$

Case 2 : 不可約二次因子

當 $F(s)$ 的分母包含一般形式的不可約二次式 $s^2 + as + b$ 時：

$$F(s) = \frac{N(s)}{(s-p)(s^2+as+b)} = \frac{A}{s-p} + \frac{Bs+C}{s^2+as+b}$$

- 求 A ：把 A 的分母遮住，即計算 $A = (s-p)F(s) \Big|_{s=p}$
- 求 B ：將等式兩邊同乘 s ，並取 $s \rightarrow \infty$ ，得到 $\lim_{s \rightarrow \infty} [sF(s)] = A + B$
- 求 C ：取簡單的 s 代入原恆等式

Question Decompose the following fraction as partial fraction.

$$\triangleright \frac{3x^2 + 7x + 28}{x(x^2 + x + 7)}.$$

$$\text{Let } \frac{3x^2 + 7x + 28}{x(x^2 + x + 7)} = \frac{A}{x} + \frac{Bx + C}{x^2 + x + 7},$$

$$\text{then } 3x^2 + 7x + 28 = A(x^2 + x + 7) + (Bx + C)x.$$

$$\text{Therefore : } \begin{cases} A = \frac{3x^2 + 7x + 28}{x^2 + x + 7} \Big|_{x=0} = 4 \\ B : \lim_{x \rightarrow \infty} [xF(x)] = 3 = A + B \implies B = -1 \\ C : \text{Set } x = 1 \implies \frac{38}{9} = 4 + \frac{-1 + C}{9} \implies C = 3 \end{cases}$$

$$\text{Hence, } \frac{3x^2 + 7x + 28}{x(x^2 + x + 7)} = \frac{4}{x} + \frac{-x + 3}{x^2 + x + 7}.$$

$$\blacktriangleright \frac{5x^2 - 2x + 7}{(x-1)(x^2 + 4)}.$$

$$\text{Let } \frac{5x^2 - 2x + 7}{(x-1)(x^2 + 4)} = \frac{A}{x-1} + \frac{Bx + C}{x^2 + 4},$$

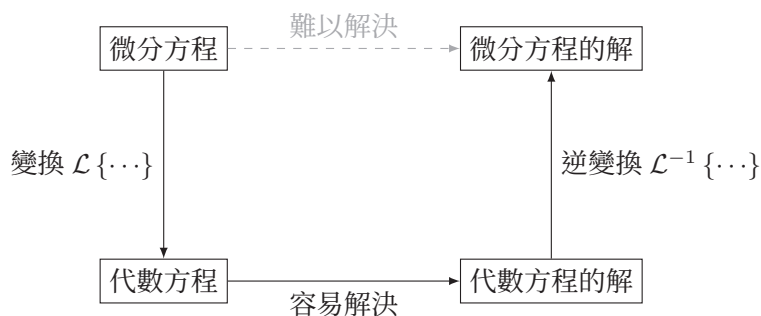
$$\text{then } 5x^2 - 2x + 7 = A(x^2 + 4) + (Bx + C)(x-1).$$

$$\text{Therefore : } \begin{cases} A = \frac{5x^2 - 2x + 7}{x^2 + 4} \Big|_{x=1} = 2 \\ B : \lim_{x \rightarrow \infty} [xF(x)] = 5 = A + B \implies B = 3 \\ C : \text{Set } x = 0 \implies \frac{7}{-4} = \frac{2}{-1} + \frac{C}{4} \implies C = 1 \end{cases}$$

$$\text{Hence, } \frac{5x^2 - 2x + 7}{(x-1)(x^2 + 4)} = \frac{2}{x-1} + \frac{3x+1}{x^2 + 4}.$$

2. 以拉普拉斯變換解微分方程

正如在 Handout7 對於拉普拉斯變換特性的介紹，它能夠將在 t -domain 的 n 階微分轉化為在 s -domain 乘上 s^n 、能夠將對於 t 的微分方程轉化為對於 s 的代數方程式。因此，當瞭解拉普拉斯逆變換時，就能夠更容易的解決微分方程：



Question Solve the following initial value problems (IVP).

▷ $y'' - 3y' + 2y = 0, \quad y(0) = 1, y'(0) = 4.$

$$\Rightarrow y'' - 3y' + 2y = 0$$

$$\Rightarrow \mathcal{L}\{y'' - 3y' + 2y\} = \mathcal{L}\{0\}$$

$$\Rightarrow (s^2 Y(s) - sy(0) - y'(0)) - 3(sY(s) - y(0)) + 2Y(s) = 0$$

$$\Rightarrow (s^2 - 3s + 2)Y(s) - s - 4 + 3 = 0$$

$$\Rightarrow (s - 1)(s - 2)Y(s) = s + 1$$

$$\Rightarrow Y(s) = \frac{s + 1}{(s - 1)(s - 2)} = \frac{-2}{s - 1} + \frac{3}{s - 2}$$

$$\Rightarrow y(t) = \mathcal{L}^{-1}\left\{\frac{-2}{s - 1} + \frac{3}{s - 2}\right\} = -2e^t + 3e^{2t}$$

► $y'' + 4y = 8, \quad y(0) = 0, y'(0) = 6.$

$$\Rightarrow y'' + 4y = 8$$

$$\Rightarrow \mathcal{L}\{y'' + 4y\} = \mathcal{L}\{8\}$$

$$\Rightarrow (s^2 Y(s) - sy(0) - y'(0)) + 4Y(s) = \frac{8}{s}$$

$$\Rightarrow (s^2 + 4)Y(s) - 6 = \frac{8}{s}$$

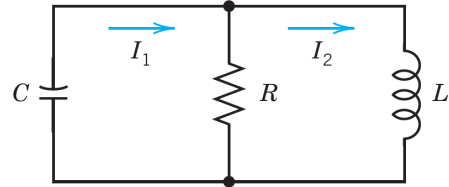
$$\Rightarrow (s^2 + 4)Y(s) = \frac{8}{s} + 6 = \frac{6s + 8}{s} = \frac{2}{s} - 2\frac{s}{s^2 + 4} + 3\frac{2}{s^2 + 4}$$

$$\Rightarrow y(t) = \mathcal{L}^{-1}\left\{\frac{2}{s} - \frac{2s}{s^2 + 4} + \frac{6}{s^2 + 4}\right\} = 2 - 2\cos(2t) + 3\sin(2t)$$

Question Solve the following IVP or BVP.

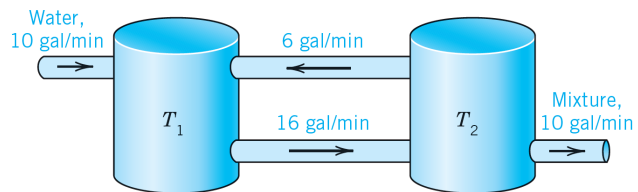
- **Circuit System.** Find the current I_1, I_2 in Fig. when $R = 1\Omega$, $L = 1.25\text{H}$, $C = 0.2\text{F}$ with initial current $I_1(0) = 1\text{A}$, $I_2(0) = 1\text{A}$.

$$\begin{aligned} & \begin{cases} 1.25I_2' + 1(I_2 - I_1) = 0 \\ 1(I_1' - I_2') + \frac{1}{0.2}I_1 = 0 \end{cases} \\ \Rightarrow & \begin{cases} I_1' + 4.2I_1 + 0.8I_2 = 0, & I_1(0) = 1 \\ -0.8I_1 + I_2' + 0.8I_2 = 0, & I_2(0) = 1 \end{cases} \\ \Rightarrow & \begin{cases} (s + 4.2)I_1 + 0.8I_2 = 1 \\ -0.8I_1 + (s + 0.8)I_2 = 1 \end{cases} \\ \Rightarrow & \begin{cases} I_1(s) = \frac{s}{(s+1)(s+4)} = \frac{-1/3}{s+1} + \frac{4/3}{s+4} \\ I_2(s) = \frac{s+5}{(s+1)(s+4)} = \frac{4/3}{s+1} - \frac{1/3}{s+4} \end{cases} \\ \Rightarrow & \begin{cases} I_1(t) = -\frac{1}{3}e^{-t} + \frac{4}{3}e^{-4t} \quad (\text{A}) \\ I_2(t) = \frac{4}{3}e^{-t} - \frac{1}{3}e^{-4t} \quad (\text{A}) \end{cases} \end{aligned}$$



- **Tank Mixing problem.** Tank T_1 in Fig. initially contains 200 gal of water in which 160 lb of salt are dissolved. Tank T_2 initially contains 100 gal of pure water. Liquid is pumped through the system as indicated, and the mixtures are kept uniform by stirring. Find the amounts of salt $y_1(t)$ and $y_2(t)$ in T_1 and T_2 , respectively.

$$\begin{aligned} & \begin{cases} \frac{dy_1}{dt} = \left(\frac{6}{100}y_2 \right) - \left(\frac{16}{200}y_1 \right) \\ \frac{dy_2}{dt} = \left(\frac{16}{200}y_1 \right) - \left(\frac{16}{100}y_2 \right) \end{cases} \\ \Rightarrow & \begin{cases} y_1' + 0.08y_1 - 0.06y_2 = 0, & y_1(0) = 160 \\ -0.08y_1 + y_2' + 0.16y_2 = 0, & y_2(0) = 0 \end{cases} \\ \Rightarrow & \begin{cases} (s + 0.08)Y_1 - 0.06Y_2 = 160 \\ -0.08Y_1 + (s + 0.16)Y_2 = 0 \end{cases} \\ \Rightarrow & \begin{cases} Y_1(s) = \frac{160(s+0.16)}{(s+0.04)(s+0.2)} = \frac{120}{s+0.04} + \frac{40}{s+0.2} \\ Y_2(s) = \frac{12.8}{(s+0.04)(s+0.2)} = \frac{80}{s+0.04} - \frac{80}{s+0.2} \end{cases} \\ \Rightarrow & \begin{cases} y_1(t) = 120e^{-0.04t} + 40e^{-0.2t} \quad (\text{lb}) \\ y_2(t) = 80e^{-0.04t} - 80e^{-0.2t} \quad (\text{lb}) \end{cases} \end{aligned}$$



級數與變換彙總

泰勒級數展開

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n \quad \text{局部上用冪級數逼近}$$

單範正交基底展開

$$\mathbf{v} = \sum_{i_1}^n \langle \mathbf{v}, \mathbf{u}_i \rangle \mathbf{u}_i \quad \text{選弦波作為基底}$$

傅立葉級數

$$f(t) = c + \sum a_n \cos \frac{n\pi t}{l} + \sum b_n \sin \frac{n\pi t}{l} \quad \text{歐拉公式轉換}$$

傅立葉級數複數型

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{i \frac{n\pi t}{l}}, \quad c_n = \frac{1}{2l} \int_{-l}^l f(t) e^{-i \frac{n\pi t}{l}} dt$$

傅立葉變換

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \quad \text{連續化}$$

拉普拉斯變換

$$F(s) = \int_0^{\infty} f(t) e^{-st} dt \quad \text{加入衰減項}$$

$\delta(t)$	1	$c_1 f(t) + c_2 g(t)$	$c_1 F(s) + c_2 G(s)$
$u(t)$	$\frac{1}{s}$	$f(t) * g(t)$	$F(s) \cdot G(s)$
t^n	$\frac{n!}{s^{n+1}}$	$e^{at} f(t)$	$F(s-a)$
e^{at}	$\frac{1}{s-a}$	$f(t+a)u(t+a)$	$e^{as} F(s)$
$\sin(at)$	$\frac{a}{s^2 + a^2}$	$\frac{d^n}{dt^n} f(t)$	$s^n F(s) - s^{n-1} f(0) \dots - f^{(n-1)}(0)$
$\cos(at)$	$\frac{s}{s^2 + a^2}$	$t^n f(t)$	$(-1)^n \frac{d^n}{ds^n} F(s)$
$\sinh(at)$	$\frac{a}{s^2 - a^2}$	$\int_0^t f(\tau) d\tau$	$\frac{1}{s} F(s)$
$\cosh(at)$	$\frac{s}{s^2 - a^2}$	$\frac{1}{t} f(t)$	$\int_s^{\infty} F(\xi) d\xi$

Next Week : Final Exam 6/8-6/12.